Signals and Systems - Lecture Notes

1. Signals and Classification

A signal is a function of one or more independent variables that carries information about the state of a physical system. Continuous-time signals are defined for every value of t; discrete-time signals are defined only at integer indices n. A signal x(t) is periodic with period T if x(t + T) = x(t) for all t, and the fundamental frequency is f0 = 1/T, measured in hertz. The angular frequency is omega0 = 2*pi*f0 radians per second.

An even signal satisfies x(t) = x(-t). An odd signal satisfies x(t) = -x(-t). Every signal can be decomposed into an even part (x(t) + x(-t))/2 and an odd part (x(t) - x(-t))/2. The unit impulse delta(t) has unit area and zero width; the unit step u(t) is 0 for t < 0 and 1 for t >= 0, and du(t)/dt = delta(t).

2. LTI Systems and Convolution

A system is linear if it obeys superposition and time-invariant if a time shift in the input produces the same shift in the output. A Linear Time-Invariant (LTI) system is completely characterised by its impulse response h(t).

The output is the convolution of the input with the impulse response: y(t) = x(t) * h(t) = integral of x(tau) h(t - tau) d(tau). In discrete time, y[n] = sum over k of x[k] h[n - k]. Convolution is commutative and associative, so systems in cascade can be combined by convolving their impulse responses h(t) = h1(t) * h2(t).

For an LTI system the output to a complex exponential e^(j*omega*t) is the same exponential scaled by the frequency response H(omega). This is the eigenfunction property and it is the reason the Fourier transform is useful.

3. Fourier Analysis

The continuous-time Fourier transform pair is X(omega) = integral x(t) e^(-j omega t) dt, and x(t) = (1/2pi) integral X(omega) e^(j omega t) d(omega). Useful pairs: a rectangular pulse of width T transforms to a sinc function T*sinc(omega*T/2), and an impulse transforms to a constant.

The Fourier series applies to periodic signals and expresses them as a sum of harmonics at integer multiples of f0. The Laplace transform generalises the Fourier transform to complex frequency s = sigma + j omega and is used to analyse systems with initial conditions and unstable responses.

4. Sampling and the Nyquist Theorem

The sampling theorem states that a bandlimited signal with maximum frequency f_max can be reconstructed exactly from its samples if the sampling rate fs is greater than 2*f_max. The rate 2*f_max is the Nyquist rate, and fs/2 is the Nyquist frequency. If fs is below the Nyquist rate, aliasing occurs and high frequencies appear as spurious low frequencies.

An anti-aliasing low-pass filter is placed before the sampler to remove components above fs/2. The reconstruction (interpolation) formula is x(t) = sum over n of x[nT] sinc((t - nT)/T), where T = 1/fs is the sampling interval.